IMEN JARRAYA

Postdoctoral Fellow RIOTU Lab

CONTACT

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Location

Riyadh, Saudi Arabia

RESEARCH FOCUS

- UAV Navigation
- GNSS-Denied Localization
- Sensor Fusion
- Battery Health Monitoring
- Energy Storage Systems
- Power Electronics
- IoT & Smart Sensing
- Embedded AI

RESEARCH OUTPUT

Q1 Journals: Satellite Navigation (IF=10.1), Energy (IF=9.4) Journal of Energy Storage (IF=9.8), Engineering Apps of AI (IF=8.0)

Conference Papers: 11+ IEEE, SMARTTECH, IREC

RESEARCH GRANTS

Awarded:

Terrain-Aided Navigation
300K SAR (2023-2026)

Proposed:

Smart Harvesting
1.45M SAR
Swarm UAV
1.36M SAR
Wearable Health
1.66M SAR
Photovoltaic Systems
1.46M SAR

RESEARCH STATEMENT

Imen Jarraya, Ph.D.

RESEARCH VISION

Developing intelligent, reliable, and energy-aware autonomous systems by integrating artificial intelligence, robotics, embedded systems, power electronics, renewable energy, and smart sensing technologies.

CURRENT RESEARCH AXES

1. Autonomous UAV Localization in GNSS-Denied Environments: Developing non-visual and terrain-aided navigation using sensor fusion, DEM/GeoTIFF information, Kalman filtering, and deep learning for robust position recovery.

2. Open-Source UAV Dataset Generation: Creating frameworks combining ROS 2, Gazebo, PX4, and terrain modeling for georeferenced UAV datasets enabling reproducible research.

3. Energy Storage & EV Systems: Studying hybrid Li-ion/supercapacitor systems, battery health monitoring (SOH), and power management strategies for electric vehicles.

4. Power Electronics & Renewable Energy: DC/DC converter design, MPPT strategies, photovoltaic systems, wind energy, and PCB design using Altium Designer.

5. Smart Sensing & IoT: Embedded sensing, IoT architectures, wearable health monitoring, precision agriculture, and intelligent automation systems.

KEY PUBLICATIONS

- CLAK: CNN-LSTM-Attention and KAN for non-visual UAV localization. **Satellite Navigation** (IF=10.1) · 2026
- Two-Stage Residual EKF using XGBoost and KAN. **Engineering Applications of AI** (IF=8.0) · 2026
- SOH-KLSTM: Hybrid KAN and LSTM for battery health. **Journal of Energy Storage** (IF=9.8) · 2025
- OS-RFODG: Open-source ROS2 framework for UAV datasets. **Results in Engineering** (IF=7.9) · 2025
- 11+ conference papers in IEEE, SMARTTECH, IREC, Mediterranean Conference, and other venues

RESEARCH METHODOLOGY

My research combines theoretical modeling, algorithm development, simulation, hardware-aware implementation, and experimental validation. I employ physics-informed reasoning, data-driven machine learning (CNN, LSTM, KAN), reproducible research practices, and embedded system constraints.

FUTURE RESEARCH AGENDA

1. Resilient UAV localization with hybrid sensor fusion | 2. AI for energy-aware autonomy | 3. Embedded AI on edge devices | 4. Wearable/IoT health monitoring | 5. Hardware-integrated research platforms

VISION 2030 ALIGNMENT

My research supports Saudi Vision 2030 through digital transformation, sustainable energy integration, economic diversification via technology innovation, and development of intelligent autonomous systems for navigation, smart mobility, agriculture, health monitoring, and cyber-physical systems.